WORKED SOLUTION

STEP I 2011, Question 1

Pure • Implicit Differentiation

> **STEP I 2011, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/11-s1-q1

The key idea

Implicit differentiation gives the gradient of each curve without solving for y. After that, every part is a matter of writing down the conditions at the meeting point (p, q): it lies on the line, it lies on the curve, and the gradients are related (equal for touching, product -1 for a normal). The gradient condition gives a simple link between p and q, and substituting that link into the other two equations pins down p in two different ways, which must agree.

Solution

Part (i)

Differentiating $\dfrac{a}{x} + \dfrac{b}{y} = 1$ with respect to x, treating y as a function of x,

$$-\frac{a}{x^2} - \frac{b}{y^2}\frac{\mathrm{d}y}{\mathrm{d}x} = 0.$$

Since $b \neq 0$ we can rearrange to get

$$\frac{\mathrm{d}y}{\mathrm{d}x} = -\frac{ay^2}{bx^2},$$

as required. (Both x and y are non-zero at every point of the curve, because the equation involves a/x and b/y.)

Equal gradients. The line $ax + by = 1$ has gradient $-\frac{a}{b}$. At (p, q) the curve has gradient $-\dfrac{aq^2}{bp^2}$, so

$$-\frac{aq^2}{bp^2} = -\frac{a}{b} \quad \Longrightarrow \quad \frac{q^2}{p^2} = 1 \quad \Longrightarrow \quad q^2 = p^2,$$

where we cancelled a/b, which is allowed because $a \neq 0$. Hence $p = \pm q$.

The two cases. The point (p, q) also satisfies

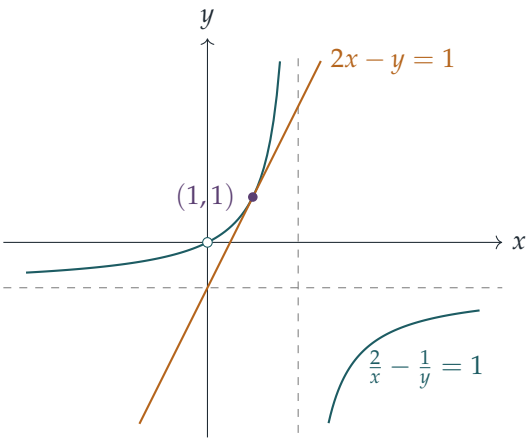
$$ap + bq = 1 \qquad \text{and} \qquad \frac{a}{p} + \frac{b}{q} = 1.$$

- If $q = p$: the line gives $p(a+b) = 1$, so $p = \dfrac{1}{a+b}$ (in particular $a + b \neq 0$). The curve gives $\dfrac{a+b}{p} = 1$, so $p = a + b$. Comparing, $a + b = \dfrac{1}{a+b}$, that is, $(a+b)^2 = 1$.
- If $q = -p$: the line gives $p(a-b) = 1$, so $p = \dfrac{1}{a-b}$. The curve gives $\dfrac{a-b}{p} = 1$, so $p = a - b$. Comparing, $(a-b)^2 = 1$.

So either $(a-b)^2 = 1$ or $(a+b)^2 = 1$, as required.

> **Ask yourself** Do such points really exist, or have we only found a necessary condition?

They do exist. Take $a = 2$, $b = -1$, so $a + b = 1$. The point $(1, 1)$ lies on the line $2x - y = 1$ and on the curve $\frac{2}{x} - \frac{1}{y} = 1$, and both have gradient 2 there. Here the curve is $y = \dfrac{x}{2-x}$, a rectangular hyperbola with the origin removed. Equal gradients at a common point mean the line is a tangent:



The open circle marks the origin, which lies on the hyperbola $y = \frac{x}{2-x}$ but not on the curve itself.

Part (ii)

Differentiating $\dfrac{a}{x} - \dfrac{b}{y} = 1$ implicitly,

$$-\frac{a}{x^2} + \frac{b}{y^2}\frac{\mathrm{d}y}{\mathrm{d}x} = 0 \quad \Longrightarrow \quad \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{ay^2}{bx^2}.$$

Suppose the line is normal to the curve at (p, q). Then $p \neq 0$ and $q \neq 0$, and

$$ap + bq = 1 \qquad \text{and} \qquad \frac{a}{p} - \frac{b}{q} = 1.$$

The tangent gradient $\dfrac{aq^2}{bp^2}$ and the line's gradient $-\dfrac{a}{b}$ are both finite and non-zero, because a, b, p, q are all non-zero. So "normal" means their product is -1:

$$\frac{aq^2}{bp^2} \cdot \left(-\frac{a}{b}\right) = -1 \quad \Longrightarrow \quad a^2q^2 = b^2p^2 \quad \Longrightarrow \quad aq = \pm bp.$$

? Ask yourself Which sign is possible?

Test each sign in the curve equation.

- If $aq = bp$, then $\dfrac{b}{q} = \dfrac{a}{p}$, so $\dfrac{a}{p} - \dfrac{b}{q} = 0 \neq 1$. This sign is impossible.
- If $aq = -bp$, then $\dfrac{b}{q} = -\dfrac{a}{p}$, so the curve equation becomes $\dfrac{2a}{p} = 1$, giving $p = 2a$, and then $q = -\dfrac{bp}{a} = -2b$.

Substituting $p = 2a$, $q = -2b$ into the line equation gives

$$2a^2 - 2b^2 = 1, \qquad \text{that is,} \qquad a^2 - b^2 = \tfrac{1}{2},$$

as required. The foot of the normal is the point $(2a, -2b)$.

Checks

- For part (i), the example $a = 2$, $b = -1$ above has $(a + b)^2 = 1$ and the line touches at $(1, 1) = (p, p)$. Similarly $a = 2$, $b = 1$ has $(a - b)^2 = 1$, and the line $2x + y = 1$ touches $\frac{2}{x} + \frac{1}{y} = 1$ at $(1, -1)$.
- For part (ii), take $a = \frac{\sqrt{3}}{2}$, $b = \frac{1}{2}$, so that $a^2 - b^2 = \frac{1}{2}$, and the point $(2a, -2b) = (\sqrt{3}, -1)$. The line gives $\frac{3}{2} - \frac{1}{2} = 1$ and the curve gives $\frac{1}{2} + \frac{1}{2} = 1$. The tangent gradient there is $\dfrac{aq^2}{bp^2} = \frac{1}{\sqrt{3}} \approx 0.577$ and the line's gradient is $-\sqrt{3} \approx -1.732$, with product -1.
- Numerically, the gradient formulae agree with finite-difference gradients at hundreds of random points, and a random search over (a, b) finds normals only when $a^2 - b^2$ is within rounding of 0.5.

Common slips

- Dropping the case $q = -p$ in part (i): it is this case that gives $(a - b)^2 = 1$.

- In part (ii), stopping at $aq = \pm bp$ without checking which sign the curve allows.

Beyond STEP

Implicit differentiation is a first glimpse of the implicit function theorem: near any point where $\partial F/\partial y \neq 0$, the equation $F(x, y) = 0$ defines y as a differentiable function of x, with $\mathrm{d}y/\mathrm{d}x = -F_x/F_y$. The tangency argument here also has an algebraic twin: a line touches a conic exactly when the quadratic for their intersections has a repeated root, and you can rederive part (i) that way. For the analysis behind all this, try Abbott, *Understanding Analysis*.

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